Raghav Senthilkumar

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Education

University of Maryland **College Park, MD**
Major: Computer Science | Minor:Economics *Anticipated May 2027*
- **GPA:** 3.92

Experience

Me Jr. **College Park, Maryland**
Database Developer *September 2024 - December 2024*
- Collaborated with a team of developers and researchers to develop an ETL process that extracted, transformed, and loaded adverse event data from FAERS and ClinicalTrials.gov into a MySQL database.
- Led the automation of generating, revising, and submitting Periodic Safety Update Reports (PSURs) for over 100 safety cases, ensuring regulatory compliance for healthcare documents.
- Presented data integration pipelines, business analysis insights, and NLP optimization strategies to stakeholders, demonstrating improved data accuracy, enhanced sentiment analysis, and more precise generation by 30%.

The First_Year Innovation and Research Experience (FIRE) **College Park, MD**
FIRE Participant, Gene Silencing *August 2023 - December 2024*
- Conducted genome editing research in *C. elegans* and tested gene silencing after promoter sequence alteration
- Oversaw data collection and analysis for 50+ genetic samples using PCR, gel electrophoresis, microscopy, and computational modeling with NetLogo and R, constructing regulatory architectures to predict gene expression patterns across 100+ simulated genomes with 70% accuracy.
- Communicated research findings on gene expression patterns, regulatory architectures, and computational models to faculty and peers.

Civil Air Patrol **Bridgewater, NJ**
Cadet Chief Master Sergeant *June 2019 - June 2023*
- Mentored a team of 20 novice cadets, providing training and support to improve their pass rates for drill tests, while also enhancing their understanding of Civil Air Patrol principles, leadership skills, and teamwork.
- Managed emergency services initiatives, organizing relief efforts and providing critical support for communities affected by natural disasters.
- Delivered presentations on leadership and emergency response strategies to cadets and senior officers.

Projects

TerpMeals
- Leveraged Python to web scrape UMD dining hall menus and images for the next seven days and employed multithreading to reduce scrape time by 94% through concurrent data extraction from multiple pages.
- Designed a data modeling strategy to structure menu data efficiently, integrating MongoDB for storage and FastAPI for serving the data, enabling the Swift application to dynamically display menu items and provide personalized recommendations based on student preferences.

Personal AI Trainer
- Streamlined a real-time workout analysis system using OpenCV and pose estimation to calculate joint angles and time under tension, storing workout data in JSON format for analysis and tracking.
- Created a RAG model using FAISS and a Hugging Face LLM to retrieve and analyze PubMed articles and real-time workout data, generating optimized workout recommendations based on scientific research.

Drug Discovery: Cyclooxygenase-2 Website
- Analyzed 1,000 drugs targeting inflammation-causing proteins from the ChEMBL database, extracting key molecular properties for predictive modeling and executing a scikit-learn machine learning model to assess drug efficacy, achieving 48% accuracy.
- Developed a website using React and Flask, integrating API calls, file manipulation, and bash commands across platforms to evaluate the efficacy of new medications for inflammation in patients.

Skills

Languages: Java, JavaScript, C#, C, Python, R, and TypeScript
Frameworks: AWS, React, REST API, SQL, Microsoft 365, PowerBI, HTML/CSS, Django, Git, and CRUD